Comparing Europe with Data

A Ulysseus COIL on comparative socio-economic data analysis

Student guide · Econometrics · Advanced Data Analysis and Visualization · 12 October – 13 December 2026
(provisional dates)

Why join?

Nine weeks, three universities, one real data project. This is what you get.

Extra points
+1 / 10 (University of Seville); +20 / 100 (Technical University of Košice · University of Montenegro). Extra points on your course grade for students who complete the final presentation.

Certificate
Certificate of participation in a Ulysseus European University COIL, signed by three European universities: Seville, Košice and Montenegro. Add it to your CV and LinkedIn.

International, from home
Nine weeks working with students from Spain, Slovakia and Montenegro, without travelling.

A real project for your portfolio
Real European data and a dashboard you can show to employers. Final projects are published in the Projects section.

Skills employers ask for
Data analysis with Power BI, Gretl, Python or R, dashboards, international teamwork and presenting in English.

A head start for your thesis
Develop your COIL project further in your bachelor's thesis (TFG), in agreement with your supervisor.

A step towards internships abroad
An international project and contacts in three countries strengthen applications such as Erasmus+ traineeships.

Your Ulysseus network
Meet students and teachers from other Ulysseus universities: a first step towards an exchange.

Teaching team

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Learning outcomes

- Collaborate effectively in virtual international teams.
- Compare economic and social phenomena across different national contexts.
- Communicate analytical findings to international audiences in English.
- Apply data analysis techniques to datasets from different countries.
- Reflect on how cultural and institutional differences influence data interpretation.

What to submit each week

> There are **three deliverables**. Each one is due on **Sunday, 23:59 CET** of its week and is submitted in this website: go to the Submit section and use the button of the deliverable. The detailed requirements are on the following pages; the other weeks are working weeks.

#	Week	Due	Deliverable	Who	Format
1	2	Sunday 25 October	**Team charter and topic proposal**	Team	One PDF, maximum 2 pages (use the templates in the Submit section)
2	6	Sunday 22 November	**Analysis notebook**	Team	Your analysis file (Python or R notebook, Gretl script or session, or Power BI file) + a PDF or HTML report with all results + your dataset (CSV or Excel)
3	9	Sunday 13 December	**Final dashboard, presentation and final reflection**	Team (dashboard, slides and presentation) + individual (reflection)	Dashboard link or PDF + slides (maximum 6 slides, 5-minute presentation) + one-page PDF reflection

What is this COIL and what will you do?

A COIL (Collaborative Online International Learning) connects your course with courses at two partner universities of the Ulysseus alliance. For nine weeks you will work online in an international team with students from Seville, Košice and Podgorica.

Together you will choose a socio-economic question, analyse real data from European countries with the methods of your course (Econometrics or Advanced Data Analysis and Visualization), build a dashboard and present your results to everybody in a joint online session. The COIL is part of your own course: you receive credits and are graded by your home university.

Before you start: checklist

- Read this guide and the schedule.
- Make sure you can use the tool of your course: Power BI, Gretl, Python or R.
- Download the COIL dataset from the Data section and open it once.
- Bookmark this website: session links, submission links, data and materials are all here.
- Check that your camera and microphone work in Microsoft Teams (sessions are Teams meetings).

Working in an international team

Spain, Slovakia and Montenegro are in the same time zone (CET), so scheduling meetings is easy. Cultural and academic habits may differ, and that is part of the learning.

- Agree on one main communication channel and answer messages within 48 hours on working days.
- Meet online at least once a week and write down what was decided and who does what.
- Use English in all team communication, so that nobody is left out.
- Rotate roles every two or three weeks: coordinator, data lead, analyst, visualisation lead and editor.
- Write a team charter in week 2: goals, roles, deadlines, how you take decisions and what you do if someone does not contribute.
- If a problem in the team cannot be solved within the team, tell your teacher early.

The project step by step

- Weeks 1–2: get to know each other, form teams and choose a question that can be answered with data from several countries (deliverable 1).
- Weeks 3–4: prepare the data and explore it. Document every variable and every change you make.
- Weeks 5–6: apply the methods of the course (regression, classification, dimensionality reduction and clustering) and interpret the results in each national context (deliverable 2).
- Weeks 7–8: turn your results into a clear dashboard for a non-specialist audience and give feedback to another team.
- Week 9: present to students and teachers of the three universities and write your final reflection (deliverable 3).

Using the data

- Start from the COIL dataset; you may add other sources if you document them.
- Always cite the source of each indicator (World Bank, IMF, Eurostat…).
- Some indicators have gaps, especially for Montenegro. Do not fill them in silently: either leave them out or explain clearly what you did.
- Check units and years before comparing countries.
- Correlation is not causation: be careful with your conclusions.

Deliverables and assessment

There are three deliverables: in week 2, week 6 and week 9 (see the Submit section and the schedule). Each university grades its own students.

The COIL gives you additional points on top of your course grade: up to 1 extra point (out of 10) at the University of Seville, and up to 20 extra points (out of 100) at the Technical University of Košice and the University of Montenegro. The extra points are awarded to students who complete the final presentation of the project in week 9.

- Each deliverable is due on Sunday at 23:59 CET of its week and is uploaded in the Submit section of this website.
- If you cannot meet a deadline, tell your team and your teacher before the deadline, not after.
- Final dashboard: an interactive link, or a PDF or screenshots if Power BI publishing is not allowed in your university account.
- The final team projects will be shown in the Projects section of this website (team name and topic only, no personal data).

Reflection

At the end of the COIL you will write a short individual reflection (part of deliverable 3). Keep some notes along the way. Some questions to guide you:

- What did you expect from working with students from other countries, and what actually happened?
- How did cultural or institutional differences change the way your team read the data?
- What would you do differently in an international team next time?
- Which skill (technical or personal) improved the most?

Academic integrity and AI tools

AI tools (ChatGPT, Copilot, Claude, Gemini…) are allowed in this COIL, on two conditions: you declare how you used them, and you review everything they produce.

- Your work must be your own and your team's. Cite all sources, data and code you reuse.
- Declare your use of AI in every deliverable: add a short note saying which tool you used, for what (e.g. debugging code, improving the English, suggesting chart types) and how you checked the result.
- You are responsible for the result: check the code, the numbers and the text that an AI tool gives you before you submit them. AI tools make mistakes and can invent data or references.
- Never present AI output as data: all figures must come from the COIL dataset or from a source you cite.
- Every team member must be able to explain the analysis and the results.

Questions and contact

For anything about grades, deadlines or your course, contact your teacher at your home university (see the Team section). For questions about the joint activities or the dataset, any teacher of the COIL can help you.

Joint sessions

There are four joint online sessions with students and teachers of the three universities, on Microsoft Teams, of 45 minutes at most. Each session is given twice, on two different days and at different times, with the same content, so that it fits the timetables of the three universities: each team attends one slot together. If you cannot attend, the recording and the slides of every session are published here afterwards. The rest of the time you work with your team: meet at least once a week, and ask your home teacher whenever you need help.

Week 1 · 45 minutes · Slot A: Date and time to be confirmed · Slot B: Date and time to be confirmed

Kick-off

Goal: Meet the people you will work with, understand how the COIL works and hold your first team meeting.

5 min	Welcome from the teachers of the three universities. What is a COIL and what is Ulysseus.
10 min	How the COIL works: this website, the schedule, three deliverables, the dataset, extra points and certificate.
25 min	First team meeting in break-out rooms (teams and their slot are announced by the teachers before the session, with students from the three universities): introduce yourself with your fact about your country, choose a communication channel and a weekly meeting time.
5 min	Questions and closing.

Before the session:

- Read the student guide and the schedule.
- Check your team and your slot (sent by your home teacher before the session).
- Think of one surprising fact or figure about the economy or society of your country.

Week 3 · 45 minutes · Slot A: Date and time to be confirmed · Slot B: Date and time to be confirmed

Data workshop

Goal: Be able to load and document the COIL data in your tool (Python, R, Gretl or Power BI) before the analysis starts.

5 min	The dataset: indicators, sources, coverage and gaps (especially Montenegro).
20 min	Live demo, about 5 minutes per tool: loading the data, selecting countries and indicators and a first chart in Python, R, Gretl and Power BI.
5 min	How to write the data dictionary and handle missing values.
15 min	Questions from the teams.

Before the session:

- Download the COIL dataset from the Data section.
- Install your tool and open the CSV once.
- Bring your team's research question.

Week 5 · 45 minutes · Slot A: Date and time to be confirmed · Slot B: Date and time to be confirmed

Mid-point clinic

Goal: Check how every team is doing and solve problems with the methods before deliverable 2 (week 6).

20 min	Lightning updates: each team presents its slide in 2 minutes.
20 min	Methods clinic in break-out rooms: regression, classification, PCA and clustering. Teachers rotate between teams.
5 min	Wrap-up: what to submit in deliverable 2.

Before the session:

- One slide per team: question, data, first result and the main problem you have.

Week 9 · 45 minutes · Slot A: Date and time to be confirmed · Slot B: Date and time to be confirmed

Final presentations

Goal: Present your project to students and teachers of the three universities. Compulsory to receive the extra points. The teachers tell each team in which slot it presents.

5 min	Opening by the teachers.
35 min	Team presentations: 5 minutes each + 2 minutes of questions and feedback (about five teams per slot).
5 min	Closing: next steps (final reflection due on Sunday, extra points and certificates) and group photo.

Before the session:

- Upload the final dashboard and the slides in the Submit section before the session.
- Rehearse: 5 minutes per team, every member presents a part.

Rules for the sessions

- The link to each session is published on this website (Sessions section and Home page). You do not need an account of another university: open the link and join from the browser or the Teams app.
- Each session is given in two slots (A and B), on different days and at different times, with the same content. Your team attends one slot together, because part of every session is work in team break-out rooms. For the kick-off and the final presentations the teachers assign the slot of each team; for the other two sessions your team chooses.
- Join on time, with your camera on if possible, and use your real name.
- The language of all sessions is English.
- The plenary parts of one of the two slots of every session are recorded, and the recording and the slides are published on this website (Sessions section) for students who could not attend. Break-out rooms are never recorded.
- If you cannot attend a session, tell your home teacher and your team in advance, and watch the recording before the next team meeting. The final presentation (week 9) is the exception: it is compulsory to receive the extra points.

Week by week: activities and deliverables

Week 1 · 12 Oct – 18 Oct · Orientation

Kick-off and intercultural orientation

Joint online welcome session. Introductions and icebreaker in mixed groups. How the COIL works.

No deliverable this week. By the end of the week: Read this website and the student guide, attend the kick-off and agree with your team on a communication channel and a weekly meeting time.

Week 2 · 19 Oct – 25 Oct · Orientation

International teams and topic selection

Teams with students from the three universities. Agree on working rules, tools and meeting times. Choose a socio-economic or public-policy question.

Deliverable 1: Team charter and topic proposal
Due: **Sunday 25 October, 23:59 CET** · Who: Team · Format: One PDF, maximum 2 pages (use the templates in the Submit section)
It must include:

- Team charter: team name, members and universities, roles, communication channel, weekly meeting time, how you take decisions, what you do if someone does not contribute.
- Topic proposal: research question, why it matters for Spain, Slovakia and Montenegro, 3–6 indicators from the COIL dataset (with their codes), countries and years, and the methods you expect to use.
- A short note on your use of AI tools (which tool, for what, and how you checked the result), or a sentence saying you did not use any.

Week 3 · 26 Oct – 1 Nov · Data

Data collection and preparation

Start from the COIL dataset; add national sources if needed. Document every variable.

No deliverable this week. By the end of the week: Your dataset is ready (one row per country and year) and every variable is documented in a data dictionary.

Week 4 · 2 Nov – 8 Nov · Data

Exploratory data analysis

Descriptive statistics, distributions, missing values, first visual comparisons of Spain, Slovakia and Montenegro within Europe.

No deliverable this week. By the end of the week: Descriptive statistics and first charts are done, and the team agrees on what the data show.

Week 5 · 9 Nov – 15 Nov · Methods

Regression and classification

Relationships between indicators across European countries and over time, and simple classification models. Interpreting results in different institutional contexts.

No deliverable this week. By the end of the week: Your regression or classification model is estimated and interpreted.

Week 6 · 16 Nov – 22 Nov · Methods

Dimensionality reduction and clustering

Principal component analysis and clustering of countries: which European countries look alike, and where do Spain, Slovakia and Montenegro fall?

Deliverable 2: Analysis notebook
Due: **Sunday 22 November, 23:59 CET** · Who: Team · Format: Your analysis file (Python or R notebook, Gretl script or session, or Power BI file) + a PDF or HTML report with all results + your dataset (CSV or Excel)
It must include:

- Data: the dataset you analysed and a data dictionary (name, code, definition, unit, source and years of every variable), with missing values and how you handled them.
- Exploratory analysis: descriptive statistics and at least three charts, one comparing Spain, Slovakia and Montenegro with the rest of Europe.
- Regression or classification: model specification, results table and interpretation in plain English.
- PCA and clustering: variance explained, number of clusters and why, and where Spain, Slovakia and Montenegro fall.
- Limitations (sample size, missing data, causality) and your three main conclusions.
- A short note on your use of AI tools (which tool, for what, and how you checked the result), or a sentence saying you did not use any.

Week 7 · 23 Nov – 29 Nov · Communication

Dashboards and data storytelling

Design the dashboard for a non-specialist audience. Give and receive feedback with another team.

No deliverable this week. By the end of the week: A first version of your dashboard, shown to another team for feedback.

Week 8 · 30 Nov – 6 Dec · Communication

Final dashboard and rehearsal

Incorporate feedback, prepare the presentation, rehearse online.

No deliverable this week. By the end of the week: Final dashboard and slides ready; the presentation has been rehearsed.

Week 9 · 7 Dec – 13 Dec · Presentation

Joint online presentation and reflection

All teams present to students and teachers from the three universities. Closing discussion.

Deliverable 3: Final dashboard, presentation and final reflection
Due: **Sunday 13 December, 23:59 CET** · Who: Team (dashboard, slides and presentation) + individual (reflection) · Format: Dashboard link or PDF + slides (maximum 6 slides, 5-minute presentation) + one-page PDF reflection
Upload the dashboard and the slides before the joint presentation. The reflection is due on Sunday.
It must include:

- Final dashboard: your question, the data, three key findings and limitations, for a non-specialist audience.
- Slides and live presentation in the joint online session: every team member presents a part. This is required to receive the extra points.
- Final individual reflection (300–400 words): what did you learn about data analysis, about working internationally and about yourself?
- A short note on your use of AI tools (which tool, for what, and how you checked the result), or a sentence saying you did not use any.

Example projects

Ideas that work with the COIL dataset. You can choose one of them or propose your own question.

Education and youth unemployment

Do countries with more young people in higher education have lower youth unemployment?

Indicators: Unemployment, youth (15-24) (SL.UEM.1524.ZS); Tertiary school enrolment (SE.TER.ENRR); GDP per capita, PPP (NY.GDP.PCAP.PP.KD)

Methods: Regression on the country panel; compare Spain, Slovakia and Montenegro with the European pattern.

The digital divide in Europe

Which European countries are digital leaders and which are lagging behind, and does it depend on income?

Indicators: Individuals using the Internet (IT.NET.USER.ZS); Fixed broadband subscriptions (IT.NET.BBND.P2); GDP per capita, PPP (NY.GDP.PCAP.PP.KD)

Methods: Correlation, then clustering of countries by their digital profile.

Ageing and public spending

Do older societies spend more on health, and do they have more public debt?

Indicators: Population aged 65 and above (SP.POP.65UP.TO.ZS); Current health expenditure (SH.XPD.CHEX.GD.ZS); General government gross debt (GGXWDG_NGDP)

Methods: Regression with time trends; discuss differences in welfare systems.

Women in the labour market

Is higher female participation in the labour market linked to fertility and income across Europe?

Indicators: Female labour force participation (SL.TLF.CACT.FE.ZS); Fertility rate (SP.DYN.TFRT.IN); GDP per capita, PPP (NY.GDP.PCAP.PP.KD)

Methods: Regression and classification of countries into high and low participation groups.

The green transition

Are richer countries greener? Renewable energy and income in Europe.

Indicators: Renewable energy consumption (EG.FEC.RNEW.ZS); GDP per capita, PPP (NY.GDP.PCAP.PP.KD); CO2 emissions per capita (excl. LULUCF) (EN.GHG.CO2.PC.CE.AR5)

Methods: Regression and clustering. Note: there are no CO2 data for Montenegro, so explain how you deal with it.

Inequality and the structure of the economy

Are economies with a larger service sector more or less unequal?

Indicators: Gini index (SI.POV.GINI); Services, value added (NV.SRV.TOTL.ZS); GDP growth (NY.GDP.MKTP.KD.ZG)

Methods: PCA to summarise the economic structure, then clustering.

Frequently asked questions

Getting started

Do I need to know how to code?

No. Everything in the COIL can be done with menus in Power BI or Gretl. Python and R are optional, for students whose course already uses them. The methods module (Resources) shows the menus first for every method, and the code only as an extra.

Which course is the COIL part of?

Econometrics or Advanced Data Analysis and Visualization, depending on your university and degree. The COIL is part of that course: you receive its credits and you are graded by your home teacher.

Which tool should I use?

The one you use in your course: Python, R, Gretl or Power BI. Students in the same team can use different tools, but agree on one for the final notebook and the dashboard.

Is my English good enough?

English is the working language of the COIL, but you do not need perfect English: clear and simple is enough. You may use AI tools to improve your English if you declare it.

Teams and topics

Who decides the teams?

The teachers, so that every team has students from the three universities. Your home teacher tells you your team and your kick-off slot before the first session (week 1).

Can we choose our own topic?

Yes. You can take one of the project ideas (Projects section), adapt it, or propose your own question in deliverable 1, as long as it can be answered with data from several countries.

Can we use data that are not in the COIL dataset?

Yes, if you cite the source and describe every variable in your data dictionary (code, unit, source and years).

What if a team member does not contribute?

Your team charter says what you do in that case. Apply it, and if the problem continues, tell your home teacher early.

We are in different countries. What about time zones?

Spain, Slovakia and Montenegro are all in the same time zone (CET), so meetings are easy to schedule.

Joint sessions

Why are there two slots for each session?

So that every student can attend despite the different timetables of the three universities. The two slots are on different days and at different times, and have the same content. Your team attends one slot together, because part of every session is work in team break-out rooms. For the kick-off and the final presentations the teachers assign the slot; for the data workshop and the mid-point clinic your team chooses.

Are the joint sessions compulsory?

The final presentation (week 9) is required to receive the extra points. The other three are strongly recommended. If you cannot attend one, tell your home teacher and your team in advance.

I cannot attend a session. Will I miss it?

No. One slot of every session is recorded and the recording and the slides are published in the Sessions section of this website, usually within two days. Watch it before your next team meeting. Break-out rooms are not recorded, so ask your team what they agreed.

Where do I find the session links?

On this website: Sessions section and Home page. Before the dates are fixed you will find a vote on the dates and times there. Each session has two links, one per slot. You do not need an account of another university to join.

Deliverables and grades

How and when do we submit?

In the Submit section of this website, with the button of each deliverable. Each deliverable is due on Sunday at 23:59 CET of its week: week 2, week 6 and week 9.

What if we cannot meet a deadline?

Tell your team and your home teacher before the deadline, not after.

Power BI does not let me publish my dashboard. What now?

Submit a PDF export or screenshots, an interactive HTML file made with Python or R, or the Gretl charts in one PDF. Any of these is fine (see Resources).

Who grades me?

Your teacher at your home university, following the rules of your own course.

How do I get the extra points and the certificate?

The extra points (+1 out of 10 at the University of Seville; +20 out of 100 at the Technical University of Košice and the University of Montenegro) go to students who complete the final presentation. You also receive a certificate of participation signed by the three universities.

AI tools and help

Can I use ChatGPT or other AI tools?

Yes, on two conditions: you declare in every deliverable which tool you used and for what, and you check everything it produces. All figures must come from the data, never from the AI.

I have another question.

Ask your home teacher (see Team). For the joint activities or the dataset, any teacher of the COIL can help you.

Templates

Template: topic proposal (week 2)

- Team name and members (name, university).
- Research question (one sentence, ending with a question mark).
- Why it matters for Spain, Slovakia and Montenegro (3–4 sentences).
- Indicators: name and code of 3–6 indicators from the COIL dataset.
- Countries and years you will use.
- Methods you expect to use.
- Possible problems (missing data, comparability) and how you will handle them.

Template: team charter (week 2)

- Our goal as a team.
- Roles for the first weeks (coordinator, data lead, analyst, visualisation lead, editor) and when we will rotate them.
- Communication channel and maximum answer time.
- Weekly meeting day and time (CET).
- How we take decisions.
- What we do if a member does not contribute or misses a deadline.

Template: data dictionary (week 3)

- One row per variable with these columns: variable name · indicator code · definition · unit · source · years covered · missing values and how they were handled.

Resources

- Ulysseus COIL Handbook: A Practical Guide for Educators: https://ulysseus.eu/wp-content/uploads/2025/02/Ulysseus-COIL-Handbook-for-teachers-1.pdf
- World Bank Open Data (source of most indicators): https://data.worldbank.org
- IMF DataMapper (government debt): https://www.imf.org/external/datamapper
- Eurostat database (more European indicators): https://ec.europa.eu/eurostat/data/database

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